Chalmers University of Technology

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A Minimal-Input Framework for Cut-In Detection and Pair-Specific Risk Analysis in Highway Trajectory Data

Traj2Rel-SFC: Trajectory-to-Relation Reconstruction and Context Signatures for Cut-In Interactions

Master's Thesis in Software Engineering (30 credits)

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Abstract

Lane ids and neighbour ids are important when studying vehicle interactions in trajectory datasets. A common limitation, however, is that many analysis pipelines depend on these derived fields directly, which reduces reuse when the fields are unavailable, inconsistent, or defined differently across datasets. This thesis presents Traj2Rel-SFC, a minimal-input framework that reconstructs lane assignment and same-lane predecessor/follower relations from trajectory geometry (x, y) and lane-marking metadata.

Based on these reconstructed relations, the framework detects cut-ins as explicit cutter–follower pairs, computes distance headway (DHW), time headway (THW), and time-to-collision (TTC) for that pair, and represents the surrounding traffic context with an ego-centric occupancy grid. To make the context comparable across travel directions, the grid is mirror-normalised and encoded as a reversible 16-bit Hilbert space-filling-curve signature.

The framework is evaluated on the *highD* dataset, where highD lane and neighbour identifiers are used only as reference labels and not as method inputs. Across recordings 01–60, same-lane predecessor/follower reconstruction with inferred lanes reaches mean accuracy 0.999810/0.999811, and cut-in events detected from reconstructed relations match the reference-label events with $F_1 = 0.998938$. The detector extracts 10,026 cut-ins, of which 10,023 remain after stage-window validity checks. For 1,421,980 stage rows, reconstructed and reference context signatures agree exactly in 99.49% of cases, and the SFC encoding is additionally verified through exhaustive round-trip testing over all 2^{16} code values. Using the mirrored decision-stage signatures, the thesis further derives seven stable context archetypes; the most risk-enriched archetype shows a 1.32× higher execution-stage THW risk prevalence than the overall event set.

The main contribution of the thesis is methodological rather than purely dataset-specific. Relation fields that are usually treated as fixed dataset columns become explicit intermediate outputs that can be reconstructed, audited, and reused for pair-consistent indicator computation and reversible context encoding. This makes the analysis reproducible, easier to inspect, and more transferable to datasets that provide trajectories and lane-boundary information but not neighbour relations directly.

Keywords: cut-in detection, minimal-input reconstruction, pair-specific risk analysis, surrogate safety measures, interaction context signature, space-filling curve, highD, reproducibility.

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We thank our supervisor for continuous feedback and for pushing us to define a clear system boundary: what information the framework is allowed to use as input, and what information must be treated as evaluation labels. We also thank the course staff and seminar group for discussions about scope, evaluation, and writing. Finally, we are grateful to the authors of the highD dataset for making the dataset available to the research community.

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Abbreviations

Abbreviation	Meaning
DHW	Distance headway (longitudinal gap)
THW	Time headway (gap divided by follower speed)
TTC	Time-to-collision (gap divided by closing speed, if closing)
SSM	Surrogate safety measure
SFC	Space-filling curve
XY-lane	Lane inference from y and lane markings

Chapter 1

Introduction

Lane changes are common on highways and are not necessarily unsafe. A *cut-in* is a lane change where a vehicle moves into a lane directly in front of another vehicle, thereby creating a direct longitudinal interaction in the target lane. Depending on available gap, relative speed, and driver behaviour, a cut-in can lead to uncomfortable or safety-critical situations.

Trajectory datasets enable large-scale analysis of such interactions. However, many datasets provide surrogate safety measures (e.g., TTC/THW/DHW) only for a *predefined* neighbour relationship (typically "ego–same-lane predecessor"). If we analyse cut-ins but reuse indicators computed for a different pair, the analysis becomes internally inconsistent even if computations are correct.

1.1 System boundary: analysing a dataset vs evaluating a method

A central writing and evaluation decision in this thesis is the **system boundary**. With a dataset such as *highD*, there are two valid perspectives:

- **Dataset analysis perspective:** treat highD's derived fields (lane ids, neighbour ids, and possibly TTC/THW/DHW) as part of the input, and use them to analyse traffic behaviour in highD.
- **Method-evaluation perspective:** treat highD's derived fields as *reference labels* , and evaluate a method that must work using only lower-level inputs such as trajectories and lane markings.

This thesis follows the **method-evaluation perspective**. Concretely, our *system under evaluation* reconstructs lane indices and neighbour relations from x, y

trajectories and lane markings only. HighD's lane and neighbour identifiers are used *only* as reference labels for evaluation and sanity checking.

1.2 Problem statement

This thesis addresses the lack of a reproducible framework that can rebuild lane and same-lane relations from minimal inputs, define cut-ins as explicit cutter–follower events, compute safety indicators for that pair, and represent the surrounding traffic scene in a compact form.

1.3 Limitations of matching highD alone

High agreement against highD is important, but it is not by itself a sufficient thesis contribution. If the method only reproduces the dataset's provided relations, the work may still appear to be a careful reconstruction exercise. The thesis therefore examines the analytical consequences of reconstructing the relevant pair explicitly. In the present case, this means defining a cut-in as a cutter–follower event, computing DHW, THW, and TTC for that pair, and showing how the interpretation changes across stages and thresholds. In this way, reconstruction accuracy is extended into a reusable analysis pipeline rather than remaining a matching exercise alone.

1.4 Key idea and novelty

The core design of Traj2Rel-SFC is to reconstruct lane and same-lane neighbour relations from minimal inputs, use those reconstructed relations to define cut-ins explicitly, and encode local context in a reusable compact form. The framework therefore has three connected parts:

1. a **trajectory-to-relation reconstruction layer** (Traj2Rel) that rebuilds lane assignment and same-lane neighbour relations from minimal geometric inputs (x, y and lane markings),

2. an **explicit interaction-pair layer** that defines a cut-in as a cutter–follower event with a reproducible onset and pair-consistent indicator computation, and

3. a **context-signature layer** that turns the surrounding traffic configuration into a compact, reversible integer code using a Hilbert space-filling-curve traversal.

The key design choice is the evaluation boundary. highD lane and neighbour identifiers are used only as reference labels and never as hidden inputs to the method. This design permits direct assessment of the minimal-input path and keeps the downstream cut-in analysis independent of dataset-specific relation columns. The individual ingredients are known in the literature; the contribution here is to integrate them into a coherent pipeline for cut-in analysis and to evaluate that pipeline explicitly against reference labels.

1.5 Goal

The goal is to build and evaluate a reproducible framework that:

1. infers lane ids from y positions and lane markings (XY-lane),
2. reconstructs same-lane preceding/following relations from geometry (x ordering),
3. detects lane changes and cut-ins and identifies the affected follower for each cut-in,
4. computes DHW/THW/TTC for the correct cutter–follower pair,
5. encodes stage-wise spatial context around cut-ins in a direction-consistent way using SFC representations,
6. evaluates reconstruction quality and event consistency against highD reference labels, and
7. makes the cut-in definition explicit through stage definitions, threshold sensitivity, and indicator-disagreement analysis.

1.6 Research questions

RQ1: How accurately can lane indices and same-lane neighbour relations be reconstructed from trajectories and lane markings, compared to highD's provided lane and neighbour identifiers?

RQ2: How can lane changes and cut-ins be detected using only reconstructed relations, how consistent are the detected cut-in events with a reference-label path that uses highD relations, and what does explicit cutter–follower pairing add to risk-feature construction?

RQ3: How can spatial context around cut-ins be encoded using SFC signatures, and how can the resulting representation support clear, direction-consistent context comparison against highD's reference relations?

1.7 Scope and delimitations

We focus on the highD dataset (highway traffic, drone-based trajectories) and offline analysis. The thesis does not aim to infer driver intent beyond observable trajectories, does not include simulation-based re-enactment, and does not claim that any single risk threshold is universally valid. Instead, the emphasis is on scenario extraction, metric correctness, validation against reference labels when applicable, and reusable feature generation.

1.8 Relation to the accepted proposal

This thesis is based on an accepted proposal for a "Framework for Cut-In Vehicle Detection and Lane-Change Risk Analysis in Naturalistic Highway Trajectories". During implementation, the scope was refined to emphasise (i) *minimal-input reconstruction* of lanes and neighbour relations, (ii) *explicit cutter–follower pairing* and pair-consistent indicator computation, (iii) *direction-normalised* spatial context encoding and verification, and (iv) a *clear separation* between event extraction and downstream risk labelling. The accepted proposal also mentioned computational efficiency, but the final thesis makes a stronger and better-supported contribution by focusing on method reuse, reference-label validation, and compact context encoding rather than on runtime benchmarking alone. The revised research questions (RQ1–RQ3) remain aligned with the proposal's core aim: reproducible detection and interpretable risk analysis at scale [1, 5].

1.9 Contributions

This thesis presents Traj2Rel-SFC, a reconstruction-and-signature framework for cut-in interactions, and makes the following contributions:

1. **Minimal-input relation reconstruction:** lane inference (XY-lane) and same-lane predecessor/follower reconstruction from x, y trajectories and lane markings, evaluated frame by frame against highD reference labels.

2. **Explicit cutter–follower cut-in definition:** cut-ins are detected as explicit interacting pairs with a reproducible onset definition (t_0), thereby making feature extraction consistent and auditable.

3. **Pair-specific indicator analysis:** DHW, THW, and TTC are computed for the actual cutter–follower pair and interpreted stage-wise, using literature-backed reasoning for why TTC and THW are useful in different parts of the interaction.

4. **Reversible context signatures:** a direction-normalised semantic occupancy grid is encoded into a reversible 16-bit Hilbert SFC signature and verified against reference occupancy built from highD neighbour relations.

5. **Clear evaluation setup and reproducible outputs:** a two-path design separates the minimal-input reconstruction path from the reference-label path, and the full run produces deterministic tables and figures that can be regenerated from the same settings.

1.10 Thesis outline

Chapter 2 introduces the dataset and core concepts. Chapter 3 presents the related work. Chapter 4 describes the research method and evaluation approach. Chapter 5 defines events, stages, indicators, and SFC signatures. Chapter 6 details the framework and validation. Chapter 7 reports results. Chapter 8 discusses limitations and implications, and Chapter 9 concludes.

Chapter 2

Background

2.1 Trajectory data and the highD dataset

We use the *highD* dataset, a drone-based collection of naturalistic highway trajectories [1]. HighD provides per-frame vehicle states (position, velocity, acceleration), lane identifiers, and neighbour identifiers such as same-lane preceding and following vehicles. The recordings are sampled at 25 Hz.

A recording is distributed as three data tables: per-frame trajectories, per-vehicle metadata, and recording-level metadata. For completeness, the corresponding file names in the released dataset are listed in Appendix A. Throughout the thesis we refer to these sources by their role (trajectories / track metadata / recording metadata) for terminological consistency.

2.2 Neighbour relations in highD

In addition to trajectories, highD provides identifiers of surrounding vehicles for each vehicle and time step (e.g., the closest vehicle ahead and behind in the same lane, and corresponding neighbours in adjacent lanes). These identifiers are convenient for analysing highD itself and, in this thesis, serve as *reference labels* for evaluating the reconstruction method. The reconstruction path described in this thesis does not use these identifiers as inputs; it derives neighbour relations from geometry and compares them against the highD-provided identifiers. A mapping between the terminology used in this thesis and the relevant highD columns is provided in Appendix A.

2.3 Rationale for pair-specific indicator computation

For cut-in risk, the key interaction is between the cutter and the follower that ends up behind the cutter in the target lane. Even if the dataset provides surrogate measures, we compute DHW/THW/TTC ourselves for the cutter–follower pair to ensure that analysis reflects the intended interaction and to support framework variants where neighbours and lanes are reconstructed from x, y and lane markings.

Chapter 3

Related Work

This chapter reviews (i) surrogate safety measures, (ii) lane-changing and cut-in analysis, (iii) highD and dataset constraints, and (iv) compact context encodings for interaction analysis.

3.1 Literature identification approach

This thesis is not presented as a full systematic literature review. However, to make the related-work selection transparent and reproducible, we followed a structured search-and-selection approach inspired by evidence-based guidelines [13, 14]. We searched IEEE Xplore, the ACM Digital Library, and Google Scholar (with Scopus used when available) for peer-reviewed work on (i) surrogate safety measures, (ii) lane changes and cut-ins, (iii) *highD* and similar trajectory datasets, and (iv) spatial/context representations for traffic interactions. The search emphasised recent work relevant to *highD* (primarily 2018–2025) while also including seminal papers needed to define key concepts.

Example search strings used during the search include:

```
("cut-in" OR "lane change") AND (trajectory OR dataset) AND
(risk OR safety)
"highD" AND ("lane change" OR "cut-in") AND
(TTC OR THW OR "surrogate safety")
("surrogate safety measure" OR "traffic conflict") AND
(validation OR review)
("occupancy grid" OR "spatial context") AND
trajectory AND vehicle
("space-filling curve" OR Hilbert OR Morton) AND
(occupancy OR grid OR encoding)
```

We included papers that address highway lane changes/cut-ins, define or review surrogate safety measures, use *highD* (or closely related datasets) for scenario extraction, or propose compact spatial context representations for interaction analysis. We complemented keyword searching with backward and forward snowballing on core papers (e.g., the *highD* dataset paper and SSM review papers) [14].

3.2 Surrogate safety measures and validation

Surrogate safety measures (SSMs) are used because crashes (collisions) are rare in naturalistic data. TTC was introduced early as a near-miss concept [2]. SSAM operationalised a set of commonly used indicators for conflict analysis and provided tooling for systematic application [3]. Extended TTC formulations have been proposed to adapt TTC-style reasoning to broader traffic conditions [4]. Recent reviews and validation-oriented studies emphasise that SSM values depend strongly on modelling assumptions, measurement noise, and traffic state, and therefore need careful interpretation [5, 6].

3.3 Lane-changing and cut-in analysis

Lane changes and cut-ins are often described through a combination of incentive, safety, and temporal structure. Classical microscopic models such as the Intelligent Driver Model (IDM) formalise desired spacing and speed adaptation in car-following [15]. Lane-changing models such as MOBIL make safety and incentive criteria explicit [16].

Naturalistic cut-in studies show that cut-ins are not just lane changes with a binary flag. Wang et al. develop an extraction algorithm for 5,608 cut-in events and report substantial variation in cut-in duration (0.7–12.4 s) together with gap-acceptance effects from environment, vehicle type, and kinematics [8]. Zhou and Itoh study driver risk perception in two lane-change stages and find that drivers tend to rely on TTC when deciding to change lanes and on both TTC and THW while cutting into the target lane [9]. This directly motivates phase-aware interpretation of surrogate measures in the present thesis.

Recent risk-prediction frameworks further reinforce the need for explicit scenario construction. On highD, Shangguan et al. integrate lane-change intention recognition with risk prediction and show that interaction characteristics with surrounding vehicles are central to risk estimation [10]. Liu et al. jointly predict lane-change manoeuvre, time-to-lane-change, and risk level, again using highD as the evaluation

basis [11]. These studies are predictive rather than reconstructive, but they support the thesis decision to preserve explicit pair identities, stage alignment, and interaction context features.

For cut-in and lane-merging behaviour in automated driving, structured phase models with explicit risk logic are also common. For example, Hwang et al. propose a finite-state-machine approach for lane-merging cut-ins with TTC-based risk handling [7]. While this thesis does not implement a controller, such phase-based framing motivates clear scenario segmentation and transparent risk definitions for offline analysis.

3.4 HighD and dataset constraints

HighD provides high-precision trajectories and neighbour relations for large-scale scenario extraction [1]. At the same time, dataset conventions and lane assignment noise can affect how lane-change timing is interpreted. Recent work discusses lane-change inconsistencies in highD and motivates careful validation when reconstructing events from lane ids and geometry [21].

3.5 Context encoding for interaction analysis

Scalar indicators (e.g., minimum THW) compress interactions to a few values and may miss relevant local traffic structure. Grid-based representations such as occupancy grids are widely used for compactly describing local surroundings [17]. For trajectory prediction and interaction modelling, pooling neighbour information into a spatial grid can improve robustness and interpretability, including for vehicle settings [18].

To compress spatial grids into a compact signature while preserving locality, space-filling curves (SFCs) such as the Hilbert curve provide a principled ordering of 2D cells into 1D sequences [19, 20]. This thesis uses a small ego-centric 3×3 occupancy grid and encodes it into a 16-bit SFC signature to obtain a reversible integer-code representation of local context.

3.6 Summary and research gap

Prior work establishes multiple surrogate safety measures, phase models, and predictive lane-change frameworks, but three gaps motivate this thesis. First, cut-in analysis requires explicit pairing between the cutter and the impacted follower;

dataset-provided SSMs may not correspond to that pair. Second, existing highD-based risk models usually assume that scenario samples and interaction partners already exist; they do not solve the minimal-input reconstruction problem when lane and neighbour identifiers are unavailable. Third, compact context representations are rarely combined with end-to-end reference-label verification. This thesis addresses these gaps via a reproducible framework that reconstructs lane and neighbour relations from minimal inputs, computes pair-consistent and phase-aware indicators, and produces verified SFC-based context signatures.

Chapter 4

Research Method

4.1 Methodological framing

This study adopts a design-oriented empirical methodology. We propose a reconstruction and representation method that derives lane indices and same-lane predecessor/follower relations directly from geometric trajectory information. The *highD* dataset is treated as a reference source for evaluation: highD-provided lane and neighbour identifiers are used as reference labels, not as inputs to the method under evaluation.

4.2 Evaluation logic: correctness and usefulness

The evaluation in this thesis serves two purposes. First, it establishes whether the minimal-input reconstruction reproduces highD's relational structure with sufficient accuracy. Second, it evaluates the analytical value of the framework once that correctness baseline has been established. For that reason, the evaluation is reported at four levels:

Frame-level validation (relations). For every frame where a relation exists, the predicted lane assignment and same-lane leader/follower identifiers are compared against the corresponding highD reference labels.

Event-level validation (cut-ins). Cut-in events detected from reconstructed relations are compared against cut-ins detected using reference relations. Agreement is reported using precision, recall, and F_1 score with an onset tolerance in frames.

Representation-level validation (occupancy and SFC signatures). For cut-in stage frames, occupancy grids are constructed both from reconstructed relations and from reference relations. The resulting binary grids and their derived SFC codes are compared using cell-wise agreement and Hamming distance. This checks whether the context signatures preserve the same neighbour-occupancy structure as the reference path.

Usefulness beyond agreement (pairing, stages, and indicators). Beyond reference-label agreement, we evaluate whether the framework leads to a clear and interpretable cut-in analysis once the interacting pair has been reconstructed. This part is reported through stage-based summaries, risk-threshold sensitivity, and TTC-versus-THW disagreement analysis. The goal is not to claim one universal threshold, but to show why explicit pair and stage definitions matter.

Qualitative verification (interactive dashboard). To complement the quantitative metrics, we performed qualitative spot checks using an interactive dashboard that synchronises video playback with reconstructed relations, event timing, and context signatures. This supported verification that the reported metrics correspond to visually plausible interactions and also provided illustrative examples.

4.3 Metrics

Lane and neighbour reconstruction are evaluated using exact-match accuracy. Because a large fraction of frames contain no neighbour (encoded as id 0), we report accuracy both over all frames and conditional on frames where the reference id is non-zero; the latter avoids inflated scores from trivial no-neighbour matches. Cut-in detection agreement is evaluated using precision, recall, and F_1 score. Occupancy agreement is evaluated using:

- **Cell-wise match ratio** for the 3×3 semantic grid:

$$\text{Match Ratio} = \frac{1}{9}\sum_{i=1}^{9} \mathbf{1}\Big(G_r[i] = G_{ref}[i]\Big),$$

  where G_r is the reconstructed grid and G_{ref} is the reference grid.

- **Hamming distance** between the 16-bit SFC codes (bitwise differences).

- **Perfect-match rate**, i.e., percentage of frames with identical grids/codes.

For the operational-usefulness layer, we additionally report event counts after acceptance criteria, risk-prevalence sensitivity under alternative THW thresholds, and TTC coverage (finite versus infinite) to show how explicit pair and stage definitions change interpretation.

4.4 Ethical and legal considerations

The thesis uses an existing trajectory dataset collected from public roads. No new data is collected and no human subjects are recruited. Derived outputs should be shared only in ways that respect the dataset licence and terms of use.

4.5 Use of Generative AI Tools

Generative AI tools (e.g., large language models) were used in a limited and supportive role during this project. AI assistance was used primarily to troubleshoot development environment configuration issues, identify and resolve Python runtime errors and dependency conflicts, and suggest refactoring patterns to standardise code structure and improve readability. AI assistance was also used to draft diagram code (e.g., Mermaid), which was then manually reviewed and exported as static figures for inclusion in the thesis.

All core research ideas, problem formulations, definitions (e.g., cut-in detection logic, stage modelling, and pair-consistent metric computation), methodological decisions, evaluation design, and interpretation of results were developed independently by the authors. AI-generated suggestions were treated as technical assistance similar to consulting documentation or community forums, and any suggested code changes were manually reviewed, validated, and tested before integration. The authors take full responsibility for the correctness of the implementation, the integrity of the analysis, and the conclusions presented in this thesis.

Chapter 5

Theory

5.1 Lane change vs cut-in

5.1.1 Lane change

We define a **lane change event** as a transition from one stable lane block to another stable lane block. "Stable" means the vehicle remains in that lane for at least a minimum number of consecutive frames before and after the transition.

5.1.2 Cut-in

We define a **cut-in event** as a lane change where the lane-changing vehicle (the *cutter*) becomes the same-lane predecessor of another vehicle (the *follower*) in the target lane. This definition yields an explicit interacting pair (A, B) with cutter A and follower B.

5.2 Cut-in onset and stage model

We define cut-in onset t_0 as the first frame where the follower's same-lane predecessor becomes the cutter id (i.e., the first frame where the follower relationship is established). Time around t_0 is segmented into four stages:

- **Intention:** $[t_0 - 4\,\mathrm{s},\ t_0 - 2\,\mathrm{s})$
- **Decision:** $[t_0 - 2\,\mathrm{s},\ t_0)$
- **Execution:** $[t_0,\ t_0 + 2\,\mathrm{s})$
- **Recovery:** $[t_0 + 2\,\mathrm{s},\ t_0 + 4\,\mathrm{s}]$

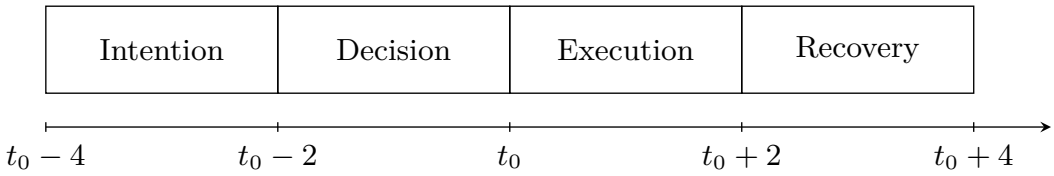

Figure 5.1: Stage model around the cut-in onset t_0 used for stage-wise feature extraction and comparison.

The four-stage decomposition is an operational definition for offline analysis rather than a claim about exact human cognitive phases. It is nevertheless consistent with structured lane-change and cut-in formulations in the literature, where intention, decision, insertion, and recovery are treated as distinct but related parts of the manoeuvre [7, 9].

5.3 Surrogate safety measures

Let A be the leader (cutter) and B the follower. We use:

- **DHW** (distance headway): longitudinal gap between leader rear and follower front.
- **THW** (time headway): $\mathrm{THW} = \mathrm{DHW}/v_B$ (with safe handling of small v_B).
- **TTC** (time-to-collision): under constant-velocity assumptions,

$$\mathrm{TTC} = \begin{cases} \mathrm{DHW}/(v_B - v_A), & \text{if } v_B > v_A \\ \infty, & \text{otherwise.} \end{cases}$$

Because cut-ins change the relevant interacting pair, all three measures are computed for the explicit cutter–follower pair rather than reusing dataset-provided values for a possibly different neighbour pair. Recent reviews emphasise that surrogate safety measures are definition-dependent and should be interpreted with care [5, 6]. In addition, Zhou and Itoh report that TTC is particularly salient to drivers when deciding whether to merge, while TTC and THW together remain informative during the actual insertion phase [9]. We therefore compute DHW, THW, and TTC throughout the stage-aligned window and interpret them as complementary rather than interchangeable indicators.

5.4 Risk label

Because risk is not standardised, we use a transparent label rule for descriptive comparison:

$$\text{risky} := \min(\text{THW}) \text{ in execution stage} < 0.7\,\text{s}.$$

This rule is not presented as a universal definition of risk. Execution-stage THW is used as this descriptive label because it remains defined after the new follower relation has been established, even when TTC becomes infinite due to non-closing conditions. The thesis therefore reports threshold sensitivity to keep the operational choice explicit rather than implicit [5, 9].

5.5 Interaction context signatures (SFC encoding)

For each frame around a cut-in event, we define an ego-centric **3×3 semantic occupancy grid** around the cutter:

- columns: {left lane, same lane, right lane},
- rows: {preceding, alongside, following}.

Each cell is treated as binary occupancy (1 if at least one vehicle occupies the cell, else 0). The 3×3 grid is embedded in a fixed 4×4 grid and encoded as a 16-bit integer using a locality-preserving space-filling curve order (Hilbert order by default). Because the mapping is deterministic and reversible, we can decode a code back into a grid.

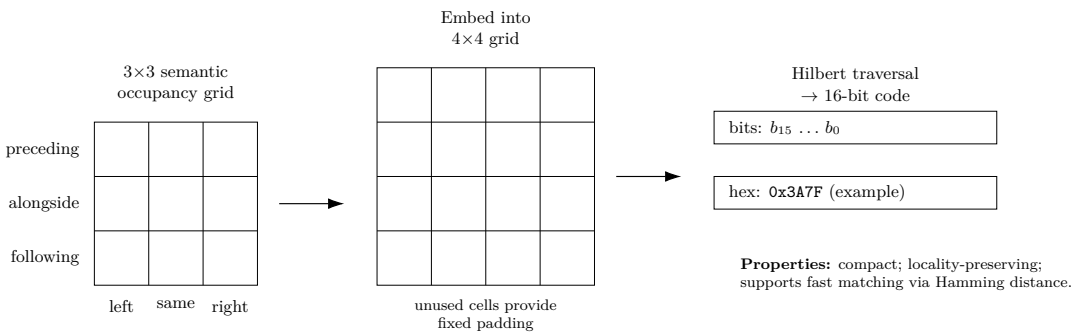

Figure 5.2: Interaction context signatures. A 3×3 ego-centric semantic occupancy grid is constructed around the cutter, mirror-normalised to a canonical direction, embedded into a fixed 4×4 grid, and encoded into a reversible 16-bit space-filling-curve (Hilbert) code.

To verify that the encoding itself is correct, we perform an exhaustive round-trip

test across all 2^{16} possible bit patterns, confirming that encoding followed by decoding returns the original grid for both Hilbert and Morton (Z-order) traversals.

Chapter 6

Methods

6.1 Framework overview

Traj2Rel-SFC is a modular framework for *trajectory-to-relation reconstruction* and *interaction-context signatures*. The core computation uses only geometric trajectory signals (x, y, velocities, vehicle dimensions) and lane-marking metadata. HighD-provided lane and neighbour identifiers are treated as **reference labels** and are used only inside evaluation components.

Two-path execution for reference-label evaluation

A central design choice is to execute the same analysis logic in two parallel paths:

- **Reference-label path:** uses highD-provided lane ids and neighbour ids to define reference relations, reference cut-in events, and reference occupancy signatures.
- **Reconstruction path (system under evaluation):** uses only x, y trajectories and lane markings to infer lane indices (XY-lane), reconstruct same-lane neighbours, and then produce the same event and signature outputs *without* using highD neighbour columns as inputs.

This design treats highD as a reference source for evaluation rather than a dependency of the method. It also enables direct, like-for-like agreement measurements at three levels: **frame-level** (lane and neighbour relations), **event-level** (cut-in detection), and **representation-level** (occupancy matrices and SFC signatures).

Macro-phases and component naming

For clarity, the framework is organised into descriptive macro-phases rather than numbered processing steps. The reconstruction path can be interpreted as a *sequence*

of deterministic processing stages: each component applies explicit acceptance criteria (e.g., lane stability, relation duration, stage-window validity) and emits traceable outputs.

Figure 6.1 illustrates the framework and the reconstruction path.

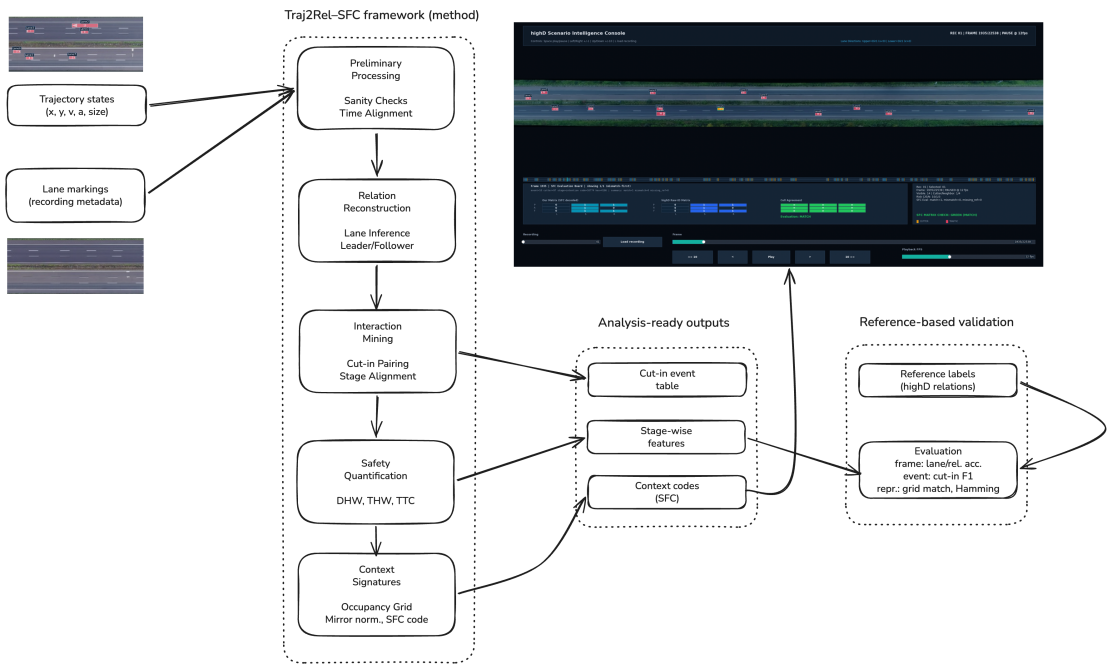

Figure 6.1: Overview of Traj2Rel-SFC and its reference-label evaluation design. The reconstruction path infers lane membership and same-lane relations from trajectories and lane markings; highD lane and neighbour identifiers are used only as reference labels for agreement measurement.

Table 6.1: Macro-phases and outputs of Traj2Rel-SFC.

Phase	Component	Main outputs
Preliminary processing	Data harmonisation	Consolidated per-frame dataset; sanity checks.
Core reconstruction	XY-lane inference (Traj2Rel)	Predicted lane index $\hat{L}$ per frame from y and lane markings.
Core reconstruction	Neighbour reconstruction (Traj2Rel)	Predicted same-lane leader and follower identifiers per frame via within-lane longitudinal ordering.
Interaction mining	Cut-in event extraction	Cut-in table with explicit cutter–follower pairs and onset t_0.
Safety quantification	Pair-consistent indicators & stages	DHW/THW/TTC for the interacting pair; stage-wise aggregates and risk summaries.
Context signatures	Occupancy & SFC encoding	Mirror-normalised occupancy matrices; reversible 16-bit SFC context signatures (hex + bits); stage-level context features.
Reference-label evaluation	Agreement reporting	Frame-level accuracy, event-level precision/recall/F_1, and representation-level match/Hamming-distance diagnostics.

The remainder of this chapter describes each component and the corresponding reference-based validation checks.

6.1.1 Qualitative inspection support

Quantitative agreement metrics provide an overall assessment of reconstruction quality, but they do not on their own reveal whether failures are systematic (e.g., due to direction conventions, onset misalignment, or lane-boundary interpretation). To support qualitative verification, we developed an interactive inspection dashboard that synchronises (i) aerial video playback and track overlays, (ii) the currently

selected cut-in event and its cutter–follower pair, and (iii) the derived stage window, indicators, and context signature for the current frame.

The dashboard was used to inspect representative cases and extremes (e.g., lowest-TTC events) and to verify that reconstructed relations and occupancy signatures correspond to the visible scene. It served as a support tool for verification and communication rather than as a standalone thesis contribution: it was not used to manually correct events or labels, and all reported results are produced by the framework's deterministic components.

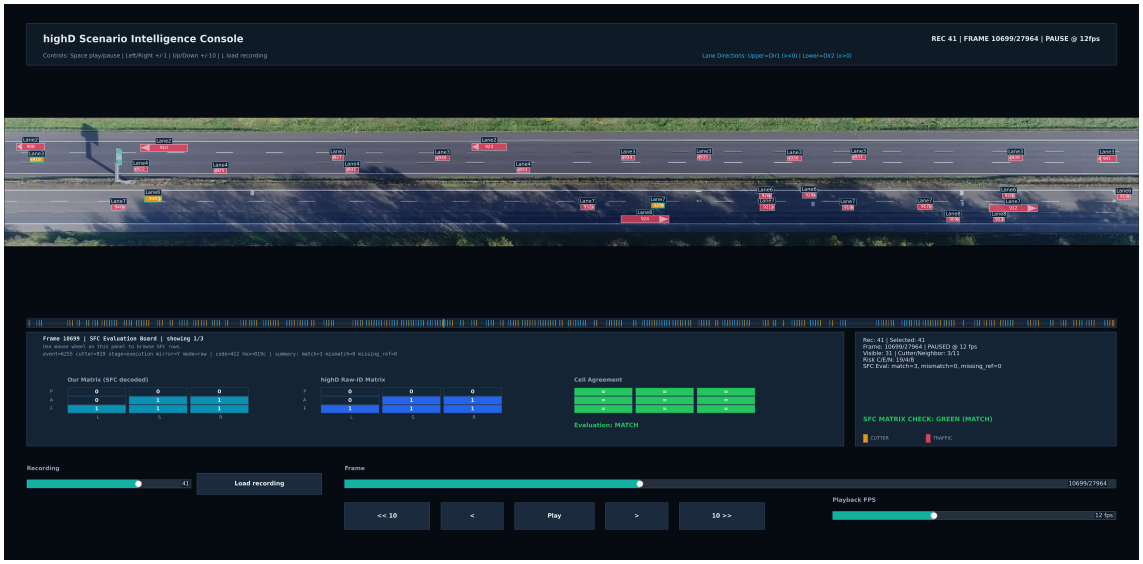

Figure 6.2: Interactive inspection dashboard used for qualitative verification and illustration. The interface links video playback with reconstructed lane and neighbour relations, highlights the cutter–follower pair for selected cut-in events, and displays the corresponding stage and context signature for the current frame.

6.2 Key configuration values

Table 6.2 summarises key configuration values used in the final run (highD 01–60, 25 Hz).

These values act as explicit cut-offs (e.g., stability requirements, search windows, and stage alignment) and are reported to support reproducibility and sensitivity analysis.

Table 6.2: Key configuration values for the final framework run.

Component	Parameter	Value
Lane-change detection	Min stable before/after	25 frames ($\approx 1.0\,\mathrm{s}$)
Lane-change detection	Ignored lane ids	$\{0\}$
Cut-in detection	Search window	50 frames ($\approx 2.0\,\mathrm{s}$)
Cut-in detection	Min relation duration	10 frames ($\approx 0.4\,\mathrm{s}$)
Cut-in detection	Max relation delay	15 frames ($\approx 0.6\,\mathrm{s}$)
Indicator computation	Min speed for THW	$0.1\,\mathrm{m/s}$
Indicator computation	Closing speed epsilon	10^{-6}
Indicator computation	Window (pre/post) for diagnostics	50/75 frames ($\approx 2\,\mathrm{s}/3\,\mathrm{s}$)
Stage model	Total analysis window	$\pm 4\,\mathrm{s}$ around t_0
Risk label	Execution THW threshold	$0.7\,\mathrm{s}$
SFC encoding	Ahead/behind range	Disabled (no fixed cutoff)
SFC encoding	Alongside threshold	$4.0\,\mathrm{m}$ (final Step 14 thesis run)
SFC encoding	Order	Hilbert

These values are operational rather than universal. Stable blocks and pair-persistence criteria make the cut-in definition explicit, while the phase windows and indicator interpretation are consistent with literature that treats lane changing as a staged process and emphasises the context-dependence of gap acceptance and risk indicators [8, 9, 7].

6.3 Preliminary processing: data harmonisation

Each highD recording is distributed as three data tables: (i) per-frame trajectories, (ii) per-vehicle metadata, and (iii) recording-level metadata. In preliminary processing we load these tables, perform basic consistency checks (e.g., frame rate and track bounds), and combine them into a consolidated per-frame dataset where each row corresponds to one vehicle at one time step together with the metadata needed in later phases. This consolidated representation is used by both the reference-label path and the reconstruction path so that subsequent comparisons are like-for-like.

We also extract lane-marking positions from the recording metadata because they provide the only map-related input used by the reconstruction method. Dataset-provided lane and neighbour identifiers are retained only for evaluation and are never used as inputs in the reconstruction path.

Algorithm 1: Lane inference from lateral position and lane markings

Input: Vehicle lateral position $y(t)$ and lane boundaries $\{b_0, \ldots, b_K\}$ from recording metadata.

Output: Inferred lane index $\ell(t) \in \{1, \ldots, K\}$ (or 0 if outside the roadway).

```
foreach frame t do
    if y(t) ∉ [b_0, b_K) then
        ℓ(t) ← 0                                  // outside valid lanes
    else
        find k such that b_{k-1} ≤ y(t) < b_k
        ℓ(t) ← k
```

6.4 Core reconstruction: lane and neighbour inference (Traj2Rel)

The reconstruction layer (Traj2Rel) derives two pieces of relational information from minimal inputs: (i) lane membership for each vehicle and time step, and (ii) the identity of the closest vehicle ahead and behind in the same lane. Together these relations are sufficient to define leader–follower interactions and to express cut-in events as explicit cutter–follower pairs.

6.4.1 Lane inference from lateral position (XY-lane)

Lane membership is inferred by mapping lateral position to the interval between two consecutive lane markings provided in the recording metadata. Values outside the roadway are mapped to lane id 0 and are excluded from interaction mining. Algorithm 1 summarises the procedure.

6.4.2 Same-lane leader/follower reconstruction

Within each frame and inferred lane, vehicles are ordered by longitudinal position in the travel direction. The immediate leader (ahead) and follower (behind) are given by adjacent vehicles in this ordering. Algorithm 2 summarises the procedure.

These two components form the core of the **system under evaluation**: starting from trajectories and lane markings, we can reconstruct the relations needed for downstream interaction mining, indicator computation, and context-signature generation without using dataset-provided neighbour identifiers as inputs.

Algorithm 2: Same-lane leader/follower reconstruction from geometry

Input: Frame-wise vehicle states with longitudinal position $x(t)$ and inferred lane index $\ell(t)$.

Output: For each vehicle: leaderId and followerId in the same lane (or 0 if none).

```
foreach frame t do
    group vehicles by lane ℓ
    foreach lane group V_{ℓ,t} do
        sort V_{ℓ,t} by x(t) in travel direction
        foreach vehicle v_i in sorted order do
            leaderId(v_i) ← closest vehicle ahead in the list (else 0)
            followerId(v_i) ← closest vehicle behind in the list (else 0)
```

6.5 Interaction mining: lane-change detection

A lane change is accepted when a vehicle transitions from one stable lane block to another, where the source lane remains unchanged for at least N_{before} frames before the transition and the target lane remains unchanged for at least N_{after} frames after the transition. Transitions involving invalid lane ids (e.g., lane id 0) are ignored.

6.6 Interaction mining: cut-in extraction (finding the follower)

Given a lane change into target lane L by vehicle A:

1. starting from the lane change start, scan frames forward within a fixed search window,

2. in each frame, identify a vehicle B in lane L for which A is the immediate leader in that frame (according to the chosen source of neighbour relations: reconstructed relations or reference labels),

3. if the relationship is established within the allowed delay and holds for at least a minimum number of frames, record a cut-in event with cutter A, follower B, and onset t_0 at the first matching frame.

Algorithm 3 summarises the rule-based cut-in identification used in the reconstruction and reference-label paths.

Algorithm 3: Cut-in identification as a cutter–follower pair

Input: Lane-change candidate for vehicle A with target lane L, reconstructed relations over time.
Output: Cut-in event (A, B, t_0) or no event.
initialise $t_0 \leftarrow$ undefined
foreach *frame t in a search window after the lane change begins* **do**
  if *A is the leader of some vehicle B in lane L* **then**
    set t_0 to the first such frame and keep B as follower
    if *the relation $A \rightarrow B$ persists for at least $d_{\min}$ frames* **then**
      return (A, B, t_0)
return no event

6.6.1 Design choices and their literature basis

The framework keeps *event extraction* separate from *risk labelling*. The settings used in the final run are design choices, but they are informed by previous work rather than selected arbitrarily.

- **Stable lane blocks.** We detect lane changes between stable lane segments rather than from single-frame label flips. This reduces sensitivity to label jitter and makes the onset definition reproducible.
- **Bounded pair formation.** A new follower must appear within a short delay and persist for a minimum duration. Naturalistic cut-in studies report wide variation in cut-in duration and gap acceptance [8], so persistence rules should be stated clearly instead of being hidden in later post-processing.
- **Pair-consistent indicator computation.** DHW, THW, and TTC are computed for the explicit cutter–follower pair, not for a generic dataset predecessor. This is important because recent lane-change risk models on *highD* rely on interaction-specific features and time-to-lane-change rather than on anonymous vehicle states [10, 11].
- **Phase-aware interpretation.** Following Zhou and Itoh, TTC is especially informative when evaluating the targeted rear vehicle before or around insertion, while THW remains informative once the pair has been established in the target lane [9]. For that reason, the thesis reports all three measures but uses execution-stage THW only as a transparent descriptive risk label.

These choices are not claimed to be universally best. Their purpose is to make the cut-in definition explicit, auditable, and amenable to later sensitivity analysis.

6.7 Safety quantification: surrogate measures and geometry model selection

We compute DHW/THW/TTC for arbitrary pairs using geometry and speeds. Because trajectory datasets may define position references differently (e.g., centre point vs bounding-box corner), we *select a geometry model* by comparing our computed indicators against highD-provided indicators for same-lane (ego, predecessor) pairs on a large random sample. We test multiple plausible reference conventions and select the one that best matches dataset-provided DHW/THW/TTC. In the final run, the best match was obtained with a bounding-box–based interpretation of the position variables (a top-left reference point), combined with vehicle length to compute front and rear positions.

To avoid negative gaps due to noise or reference mismatches, thesis-facing minima are clipped to be non-negative for DHW/THW. TTC is treated as infinite when the follower is not closing ($v_B \leq v_A$).

6.8 Safety quantification: stage-wise features

For each detected cut-in, we align a fixed time window around onset t_0 and compute stage-wise summaries for the four stages in Chapter 5. The feature table includes stage minima of DHW/THW/TTC and kinematic aggregates for cutter and follower. These stage-wise summaries are later used not only for descriptive risk analysis, but also for operational-usefulness evaluation through threshold sensitivity and TTC-versus-THW disagreement analysis.

6.9 Context signatures: encoding and reference verification

To capture local traffic context around the cutter compactly, we encode neighbourhood occupancy into the 3×3 semantic grid defined in Chapter 5. Each frame is mapped to a binary occupancy pattern, and the 3×3 cells are embedded into a 4×4 grid and linearised using a Hilbert-type space-filling-curve order. To remove direction-induced asymmetry, events are mirror-normalised to a canonical orientation prior to aggregation.

Algorithm 4: Ego-centric occupancy grid and space-filling-curve context signature

Input: Cut-in event (A, B, t_0), reconstructed lane/relations, neighbourhood range parameters.

Output: A stage-aligned sequence of binary grids and a 16-bit context signature per frame.

foreach *frame t in the analysis window around t_0* **do**

- build a 3×3 ego-centric occupancy grid around the cutter A
- (optional) mirror-normalise the grid to a canonical direction
- embed the 3×3 grid into a 4×4 grid with fixed padding
- traverse cells in Hilbert order and write occupancy bits into a 16-bit code
- store the code and (optionally) the decoded grid for verification

Minimal-input SFC signatures. In the minimal-input path, occupancy is computed from inferred lanes (XY-lane) and lane-snapshot neighbour queries (geometry). This means SFC signatures can be generated without accessing highD neighbour columns.

Reference SFC signatures. For verification, we also construct a reference occupancy grid from the highD surrounding-vehicle identifiers (same-lane preceding/following plus left/right neighbours and alongside vehicles). Mapping each non-zero neighbour id to occupancy yields a reference 3×3 grid per frame, which can be encoded into a reference SFC signature.

Signature agreement. We verify the SFC branch by comparing the reference and minimal-input codes for matching (event, frame) rows. We report both exact code match rate and the distribution of Hamming distances (bitwise differences), which localises whether mismatches affect a single cell or multiple cells.

Algorithm 4 summarises how the context signatures are constructed for each stage-aligned frame.

6.10 Post-processing: reproducible outputs

The framework is parameterised through a small set of explicit settings (e.g., stability thresholds and stage windows). Each run produces versioned outputs (tables and figures) that can be regenerated from the same raw data and settings. The thesis-facing claims in Chapters 7–9 rely on the all-recordings reconstruction, stage-feature, and context-signature artifacts; exploratory predictive experiments included elsewhere in the repository are separate and are not used as evidence for the thesis claims.

The intended entry point for reproducing the reported tables and figures is the project's reproducibility package or repository: https://github.com/arsulesandy/cutin-risk-analysis.

Chapter 7

Results

7.1 Detected events and analysis coverage

The final run uses recordings 01–60. Table 7.1 summarises event yield.

Table 7.1: Event counts for the final thesis run (recordings 01–60).

Artefact	Lane changes	Cut -ins	Notes
Interaction mining (stable blocks + follower pairing)	11,856	10,026	Lane changes and cut-ins extracted from reconstructed relations.
Stage feature table (intention–recovery window)	–	10,023	Three events removed by stage-window validity checks.
Context signatures (frame-level occupancy)	–	1,421,980	Frame-level context signatures for the 10,023 validated cut-in events.
Recordings processed		60 (01–60)	

Throughout the remainder of the thesis, we distinguish clearly between *extracted cut-ins* (10,026) and *validated analysis events* (10,023 after stage-window checks). This distinction is important because the framework exposes each acceptance criterion explicitly rather than implicitly excluding events in downstream scripts.

7.2 Lane-change and cut-in duration statistics

Table 7.2 summarises dataset-wide duration statistics. Lane-change durations are computed from stable-block transitions. Cut-in relation duration is measured as the contiguous duration where the follower's predecessor is the cutter *within the*

configured search window. In our configuration, the relation duration is capped at 50 frames (2 s), which explains the concentration at the cap.

Table 7.2: Dataset-wide duration statistics (highD 01–60, 25 Hz).

Event type	Min (s)	P25 (s)	Median (s)	Mean (s)	P75 (s)	Max (s)
Lane change duration	0.960	3.680	6.240	6.551	8.960	69.120
Cut-in relation duration	0.360	1.960	1.960	1.870	1.960	1.960

Overall, 84.56% of detected lane changes are classified as cut-ins under our definition (10,026 cut-ins out of 11,856 lane changes). This proportion should be read as the yield of the operational event definition used in this thesis, not as a general prevalence estimate for highway traffic.

7.3 Evaluation of the minimal-input reconstruction (RQ1/RQ2)

Same-lane neighbour reconstruction. Geometry-based same-lane neighbour reconstruction matches the reference preceding and following relations with mean accuracy of 0.999997 across the 60 recordings. Unless stated otherwise, accuracies are computed over all frames; conditional accuracies restricted to frames where the reference relation exists (non-zero id) are reported in the supplementary material.

XY-lane variant. When lane membership is also inferred from geometry (XY-lane), agreement remains high with mean preceding and following accuracies of 0.999810 and 0.999811. This corresponds to a frame-level mismatch rate of about 0.02%. Cut-in detection based on reconstructed relations matches the reference-label events with $F_1 = 0.998938$. Across recordings, the per-recording means are tightly concentrated: the approximate 95% confidence interval over recording means is $\pm 2.3 \times 10^{-5}$ for preceding/following accuracy and $\pm 7.3 \times 10^{-4}$ for cut-in F_1.

These results answer the correctness question: the minimal-input path reproduces the reference-label path with high agreement. The remaining results address usefulness: how explicit pair construction, stage alignment, and explicit indicator choices affect interpretation once this correctness baseline is established.

7.4 Execution-stage indicator distributions

Table 7.3 summarises execution-stage minima across all 10,023 events. TTC statistics are shown only for events with finite TTC (i.e., closing speed).

Table 7.3: Execution-stage indicator minima across all events (10,023 cut-ins).

Indicator	Min	P25	Median	Mean	P75	Max
$\mathrm{DHW}_{\mathrm{min}}$ (m)	1.37	20.69	32.21	51.28	59.82	373.98
$\mathrm{THW}_{\mathrm{min}}$ (s)	0.051	0.769	1.194	1.700	2.028	12.383
$\mathrm{TTC}_{\mathrm{min}}$ (s), finite only (n=3,606)	0.220	10.769	19.726	74.161	42.025	14428.50

The TTC distribution is strongly right-skewed, so the median and quartiles are more informative than the mean for interpreting the typical severity of finite-closing events.

7.5 Stage-based risk summaries

Risk is defined as:

$$\text{risky} := \min(\mathrm{THW})_{\text{execution}} < 0.7\,\mathrm{s}.$$

Using this rule on the merged stage-feature table (10,023 events), 2,018 events are classified as risky (20.1%). Table 7.4 reports pooled comparison statistics. Here Δv is defined per event as the difference between the cutter and follower mean speeds in the execution stage, i.e., $\Delta v = \bar{v}_{\text{cutter}} - \bar{v}_{\text{follower}}$ (reported as the median across events).

Table 7.4: Execution-stage pooled risk summary (all recordings). Δv is in m/s.

Group	Events	$\mathrm{THW}_{\mathrm{min}}$ median (s)	$\mathrm{DHW}_{\mathrm{min}}$ median (m)	Δv median
Non-risky	8,005	1.427	39.680	2.304
Risky	2,018	0.539	14.945	2.853

7.6 Effect size comparison between risk groups

Because the dataset contains over 10,000 cut-in events, even small group differences can appear statistically significant under conventional hypothesis tests. To describe *magnitude* rather than only significance, Table 7.5 reports Cliff's delta (δ) for selected execution-stage variables. Cliff's δ ranges from -1 (all risky values smaller than

non-risky) to +1 (all risky values larger), and values near 0 indicate substantial overlap between the groups.

Table 7.5: Effect-size comparison between risk groups (execution stage). Medians are reported per group; δ is Cliff's delta (risky vs non-risky).

Metric	Median non-risky	Median risky	Cliff's δ
$\mathrm{DHW}_{\mathrm{min}}$ (m)	39.68	14.95	−0.950
Δv (m/s)	2.304	2.853	0.032
$\lvert v_{y,\mathrm{cutter}}\rvert_{\mathrm{max}}$ (m/s)	0.88	0.93	0.097

The separation in $\mathrm{DHW}_{\mathrm{min}}$ is expected because THW is directly related to spacing via $\mathrm{THW} = \mathrm{DHW}/v_B$. In contrast, the lateral-velocity difference is modest, indicating that the THW-based label primarily captures longitudinal gap acceptance rather than the lateral kinematics of the lane change.

7.7 Sensitivity of the THW threshold

To illustrate that the risk label is a design choice, Table 7.6 shows how the risk prevalence changes with the THW threshold.

Table 7.6: Sensitivity of risk prevalence to the THW execution-stage threshold (10,023 events).

THW threshold (s)	Risky events	Risk ratio
0.50	760	7.58%
0.70	2,018	20.13%
1.00	4,004	39.95%
1.50	6,235	62.21%

7.8 Indicator disagreement example: THW vs TTC

TTC is only finite when the follower is closing in on the cutter ($v_B > v_A$). In this dataset, execution-stage TTC is finite for 3,606 of 10,023 events (35.98%). Within this finite subset, risky events (by THW) have substantially lower TTC.

Table 7.7: Execution-stage TTC for events with finite TTC only (3,606 events).

Group (by THW label)	Events	TTC median (s)	TTC mean (s)
Non-risky	3,177	22.30	80.27
Risky	429	8.98	28.94

Table 7.8 compares the THW-based label to TTC thresholds over all events, treating infinite TTC as "not risky". Strict TTC thresholds have high precision but low recall relative to the THW label because many THW-risky events do not involve a closing-speed situation.

Table 7.8: Overlap between THW-risky and TTC-threshold flags (all events; infinite TTC counted as not risky).

TTC thr. (s)	TTC-flagged	TP	FP	Prec.	Rec.
1.5	10	10	0	1.000	0.005
3.0	43	37	6	0.860	0.018
5.0	156	99	57	0.635	0.049
10.0	788	237	551	0.301	0.117

7.9 Importance of explicit pairing and phase-aware measures

The high agreement reported above addresses the correctness question, but it does not by itself establish the analytical value of the framework. That value arises from making the event definition and the indicator interpretation explicit.

First, the event pipeline is traceable from end to end: 10,026 cut-ins were extracted, 10,023 remained after stage-window validity checks, and the stage feature table keeps the cutter and follower identities throughout the aligned window. This means the acceptance rules are visible in the method instead of being hidden inside dataset-specific scripts.

Second, Table 7.6 and Table 7.7 show that one indicator cannot explain all phases equally well. When the execution-stage THW threshold is changed from 0.50 to 1.50 s, the risky-event share changes from 7.58% to 62.21%. At the same time, TTC is finite in only 35.98% of execution-stage events. A TTC-only post-merge analysis would therefore ignore most established cut-ins even though the interacting pair is clearly defined.

This pattern is consistent with previous work showing that lane-change risk perception is stage dependent and that TTC and THW describe different parts of

the interaction [9, 5]. In that sense, the contribution of Traj2Rel-SFC is not a claim that one threshold is universally correct. The contribution is that the framework reconstructs the relevant interaction pair, computes the indicators for that pair, and keeps the later label choice explicit.

7.10 Extreme events by TTC

Table 7.9 lists the three events with the lowest execution-stage TTC (finite only), which are useful for qualitative inspection using the dashboard in Figure 6.2.

Table 7.9: Top-3 extreme events by minimum execution-stage TTC (finite only).

Rec.	Cutter	Follower	From	To	TTC_{min} (s)	THW_{min} (s)
40	682	687	1	2	0.220	0.051
26	496	522	1	2	0.235	0.122
30	839	852	1	2	0.288	0.067

7.11 Per-recording variation

Across the 60 recordings, the risk ratio (THW < 0.7 s) varies between 11.1% and 34.1% (median 20.0%). This suggests that traffic composition and local conditions differ across recordings even within the same dataset.

7.12 Context signature maps

To interpret the context signatures, we report mean occupancy probabilities in the semantic 3×3 grid described in Chapter 5. The grid is aligned with the cutter after mirror normalisation; columns correspond to {left lane, same lane, right lane} and rows to {preceding, alongside, following}. The centre cell (same-lane/alongside) is occupied by construction because it represents the cutter; occupancy in other cells indicates the presence of at least one surrounding vehicle in the corresponding relative region.

We present the decision stage as a representative example because it captures the traffic configuration immediately before the cut-in interaction becomes established. Figure 7.1 shows mean occupancy for non-risk and risky events and the risk-minus-non-risk difference for the decision stage.

To highlight how group differences evolve over time, Figure 7.2 shows risk-minus-non-risk difference maps for the intention, decision, and execution stages using a

consistent colour scale. Because the differences are small in magnitude, they may appear visually subtle; supplementary absolute maps for the intention and execution stages are provided in Appendix C.

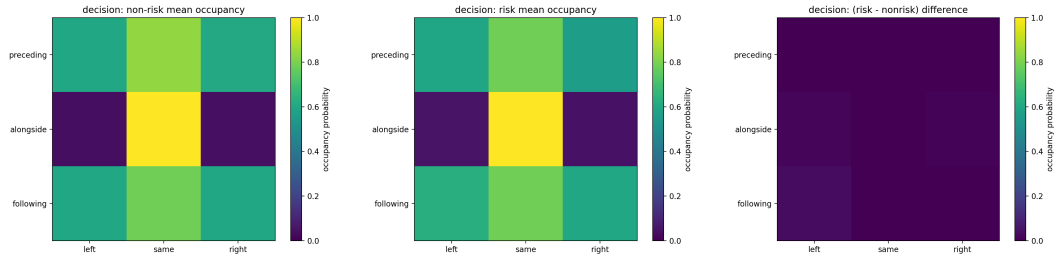

Figure 7.1: Decision-stage occupancy maps: non-risk (left), risk (middle), and risk-minus-non-risk difference (right).

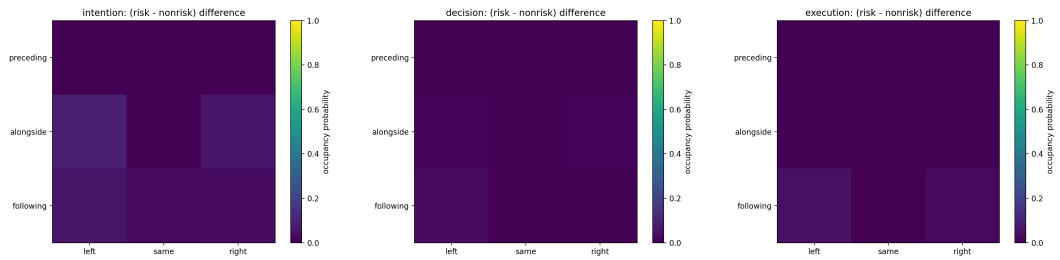

Figure 7.2: Stage-wise risk-minus-non-risk occupancy differences for intention, decision, and execution stages.

7.13 Decision-stage context archetypes

Mean occupancy maps show group-level differences, but they do not directly answer whether the verified SFC representation can recover a small set of reusable scene families. To test that, we decode the mirrored binary SFC signatures back into the semantic 3×3 grid, average the decision-stage grids within each event, and cluster the resulting event-level context vectors. We use the *decision* stage rather than the execution stage so that clustering is based on pre-outcome context rather than on the same time window used to define the THW risk label.

Searching $K = 3, \ldots, 8$ with k-means, the best solution is $K = 7$, with silhouette score 0.498 and mean stability ARI 0.992 across repeated initialisations. The resulting archetypes remain reasonably balanced, spanning 6.0%–30.7% of all 10,023 validated cut-ins. After mirror normalisation, all lane changes are represented in a common canonical direction, so the right column in Figure 7.3 corresponds to the target lane for interpretation.

Figure 7.3 shows the seven archetypes, and Figure 7.4 compares their execution-stage THW risk prevalence. The most risk-enriched archetype (A4, 10.27% of events) reaches 26.63% risky outcomes versus 20.13% overall (risk ratio 1.32) and has median execution THW 0.975 s. Its mean grid is dominated by occupied target-lane preceding and following cells, indicating insertion into a relatively constrained target-lane corridor. Lower-risk archetypes such as A2 and A5 remain below the overall prevalence (17.18% and 17.51%, respectively). This result shows that the reversible SFC code is not only verifiable against reference labels, but also rich enough to support interpretable unsupervised grouping of cut-in scenes before outcome labels are applied.

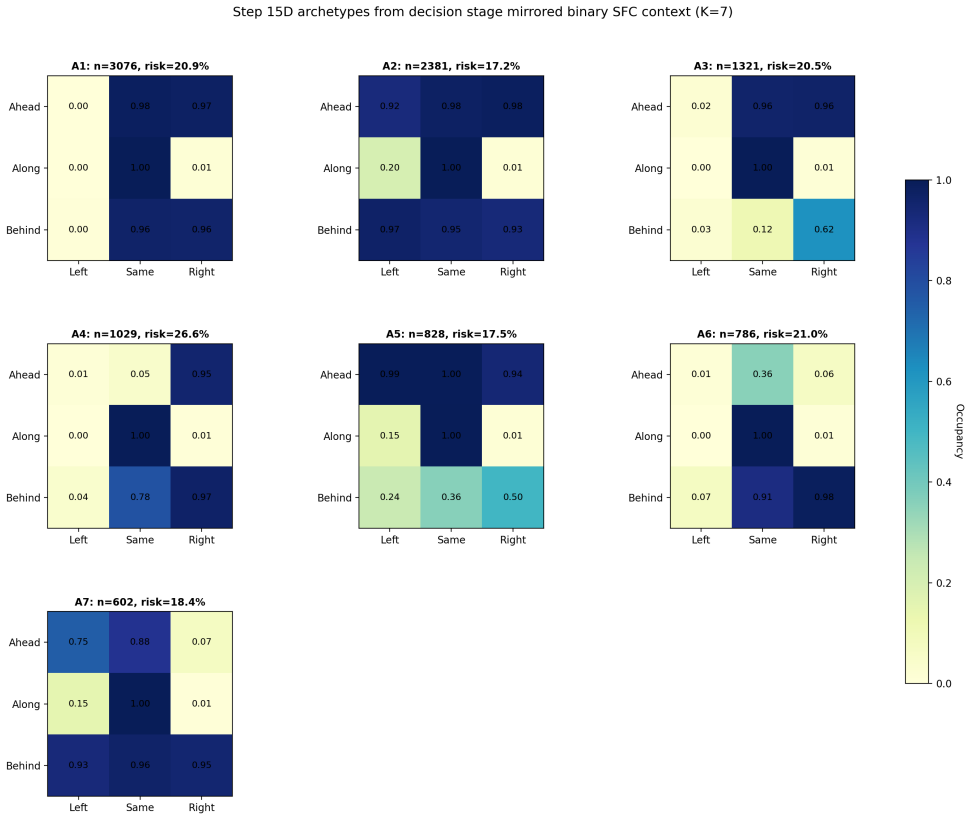

Figure 7.3: Seven decision-stage context archetypes derived from mirrored binary SFC signatures. After mirror normalisation, the right column corresponds to the target lane, so each panel visualises one canonical cut-in scene family.

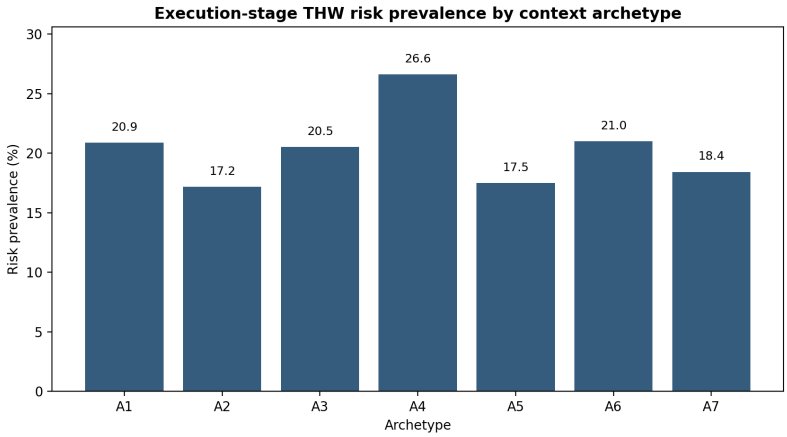

Figure 7.4: Execution-stage THW risk prevalence by decision-stage context archetype.

7.14 Reference agreement for context signatures (RQ3)

We verify the SFC encoding in two complementary ways:

- **Algorithmic round-trip:** exhaustive decode↔encode verification across all 2^{16} codes confirms that the 16-bit SFC mapping is reversible for both Hilbert and Morton orders.
- **Reference agreement:** for cut-in stage frames, we compute a reference occupancy grid from highD surrounding-vehicle ids and compare its SFC code to the minimal-input code computed from inferred lanes and geometry. Exact matches indicate identical neighbour occupancy under the two definitions; non-matches localise differences to specific cells via Hamming distance.

Using the final thesis configuration, with an alongside threshold of 4.0 m and no fixed ahead/behind cutoff, we verify 1,421,980 stage rows and obtain 1,414,762 exact row matches (99.49%). Stage-wise exact-match rates are 99.35% (intention), 99.52% (decision), and 99.60% (execution). In the thesis event view, stage-window ANY-frame agreement is 100.00%, while stage-window ALL-frame agreement is 79.82%. The raw unmirrored match rate is only 57.56%, which confirms that direction normalisation is necessary for like-for-like left/right occupancy comparison across opposite travel directions.

Chapter 8

Discussion

8.1 Answers to the research questions

RQ1 (reconstruction accuracy): Same-lane neighbour reconstruction from geometry reaches near-perfect agreement with highD reference neighbours. With inferred lanes (XY-lane), the agreement remains above 0.9998 for both preceding and following.

RQ2 (cut-in consistency): Stable-block lane-change detection combined with follower pairing yields 10,026 extracted cut-ins from 11,856 lane changes across recordings 01–60, of which 10,023 remain after stage-window validity checks. When the full cut-in detector is run end-to-end on reconstructed relations, event matching against the reference-label path achieves $F_1 = 0.998938$. Beyond correctness, the explicit cutter–follower representation enables pair-consistent surrogate-measure computation and preserves an explicit event definition.

RQ3 (SFC encoding and verification): A compact 3×3 occupancy grid encoded into a 16-bit SFC signature provides an interpretable local context representation. The encoding is reversible by design and is validated by exhaustive round-trip checks. In the final configuration, reference-vs-minimal-input matching reaches 99.49% exact row agreement over 1,421,980 verified rows (with 100% event-level ANY-frame agreement), confirming that the reconstructed occupancy used for SFC encoding corresponds closely to the reference relations in highD once direction is normalised. The same representation also supports stable unsupervised grouping: decision-stage context clustering yields seven archetypes, with the most constrained target-lane corridor archetype showing a 1.32× higher execution-stage THW risk prevalence than the overall set.

8.2 Interpretation and implications

The principal outcome of the thesis is methodological. The framework defines a clear system boundary: trajectories and lane markings serve as inputs, whereas highD relation fields serve as evaluation labels. This separation allows disagreements with dataset conventions to be analysed explicitly rather than inherited as hidden assumptions.

The innovative step is relation-first analysis: lane ids and neighbour ids are treated as reconstructable intermediate products rather than immutable dataset facts. Although algorithmically modest, this shift is scientifically important because it separates observation, derived relation, event definition, and risk label into auditable layers. The result is a pipeline whose assumptions can be inspected, challenged, and transferred more readily than those of a pipeline that starts from hidden dataset-specific relation columns.

This transferability still has conditions. The current method assumes trajectories are expressed in a consistent coordinate system and that lane-boundary metadata, or an equivalent road model, is available. The contribution is therefore not that the framework is assumption free, but that its assumptions are explicit and that their consequences can be measured against reference labels.

Table 8.1: Main contribution claims and the evidence provided in this thesis.

Contribution claim	Evidence in this thesis	Why it matters
Minimal-input relation reconstruction	Mean preceding/following accuracy 0.999810/0.999811; cut-in $F_1 = 0.998938$	Makes cut-in mining reusable on data sources that lack neighbour ids.
Explicit cutter–follower pairing	10,026 extracted cut-ins; 10,023 validated analysis events with stage-aligned pair features	Avoids reusing dataset indicators for the wrong interaction pair.
Phase-aware indicator analysis	TTC finite in 35.98% of execution-stage events; THW sensitivity changes risk prevalence from 7.58% to 62.21%	Shows that risk interpretation depends on explicit pair and stage definitions rather than on one universal threshold.
Verified context signatures	Reversible 16-bit Hilbert code; 99.49% exact row agreement over 1,421,980 stage rows; 7 stable decision-stage archetypes with peak risk ratio 1.32	Provides compact, reversible context descriptors that support interpretable grouping of cut-in scenes rather than only average occupancy plots.
Clear evaluation boundary	Two-path design, parameter table, and deterministic exports	Separates method inputs from reference labels and makes the evaluation reproducible.

Explicit pairing matters because the safety-relevant interaction after a cut-in is the cutter and the new follower in the target lane. Dataset-provided indicators computed for a generic predecessor relation may be consistent with dataset conventions yet still correspond to a different interaction pair in a cut-in study. Traj2Rel-SFC addresses this by defining cut-ins as explicit cutter–follower events and computing DHW, THW, and TTC for that pair only.

The phase-aware indicator results matter for the same reason. TTC and THW are not interchangeable: TTC focuses on closing-speed conflict, while THW focuses on accepted gap. In our data, TTC is finite for only 35.98% of execution-stage events, which means that a TTC-only post-insertion analysis would miss the majority of established cut-ins. This matches literature that treats lane-change risk perception as phase dependent [9] and broader reviews that warn against one-size-fits-all interpretation of surrogate measures [5]. The contribution here is therefore not a claim of one universally correct threshold, but a method that makes the indicator choice explicit and ties it to the correct interacting pair.

The SFC signatures complement the scalar indicators. Because the code is

reversible, it does not replace interpretation with a black-box embedding. Instead, it provides a compact summary that can be decoded, checked, and then used for large-scale comparison, retrieval, or clustering of interaction patterns. The archetype analysis makes this concrete: a pre-outcome decision-stage taxonomy can be learned directly from the verified signatures, and the resulting scene families are meaningfully associated with later execution-stage THW risk.

8.3 Threats to validity

Construct validity: Risk is defined via a THW threshold in the execution stage. Alternative definitions (e.g., TTC thresholds, required deceleration / braking-demand indicators, or post-encroachment measures) would label different subsets of events, so the reported "risk ratio" should be interpreted as *prevalence under this definition*. More generally, our criteria should be read as a stricter, pair-consistent definition of cut-in relevance, not as proof that highD is absolutely "wrong". This matters especially because highD lacks wider upstream/downstream context that may influence discretionary lane changes [21].

Internal validity: Reconstruction and metric computations depend on lane markings and coordinate conventions (e.g., bounding-box reference points). Errors in these inputs can shift onset t_0 and bias DHW/THW/TTC. We mitigate this via (i) geometry model selection against dataset-provided indicators where applicable, and (ii) independent reconstruction variants (geometry-only neighbours and XY-lane inference) that agree closely with the reference-label path.

External validity: Results are based on *highD* highway recordings and may not transfer directly to urban driving, ramp merges, or different sensing setups. Differences in speed limits, lane geometry, and traffic composition can change both cut-in dynamics and the interpretation of surrogate measures.

Conclusion validity: We report descriptive differences, sensitivity checks, and agreement checks against reference labels. The thesis does not make causal claims about driver behaviour or safety outcomes, and statistical significance is not emphasised over effect magnitude.

8.4 Future work

Natural next steps include (i) extending the surrogate-measure layer with braking-demand or minimum-safe-deceleration criteria, which have been proposed as complementary lane-change safety indicators [12], (ii) adding broader ablation studies

over relevance-distance, pair-persistence, and stage-boundary criteria, ideally with manual review of disagreement cases, and (iii) testing whether the learned context archetypes transfer across datasets, road classes, or prediction tasks rather than remaining specific to *highD*'s highway setting.

Chapter 9

Conclusion

This thesis presents Traj2Rel-SFC, a minimal-input framework for reconstructing lane and same-lane relations, detecting cut-ins as explicit cutter–follower events, computing pair-specific safety indicators, and encoding local traffic context as reversible SFC signatures. By using highD lane and neighbour ids only as reference labels, the thesis evaluates the method itself rather than simply analysing a dataset through its own derived fields.

On highD recordings 01–60, the framework reaches mean preceding/following accuracy 0.999810/0.999811 and event-level cut-in matching of $F_1 = 0.998938$. The detector extracts 10,026 cut-ins, of which 10,023 remain after stage-window checks. For 1,421,980 stage rows, the reconstructed and reference context signatures match exactly in 99.49% of cases, and the 16-bit SFC encoding is verified by exhaustive encode↔decode round-trip checks.

The thesis also demonstrates that the contribution extends beyond reconstruction accuracy. Once the cutter and follower are made explicit, DHW, THW, and TTC can be computed for the correct pair and interpreted in a stage-based way. The threshold sensitivity analysis and TTC coverage results show that subsequent risk interpretation depends strongly on indicator choice and phase.

Overall, the contribution is a clear and reusable pipeline for cut-in analysis: minimal inputs, explicit interacting pair, verified context representation, and an evaluation setup that makes each step auditable and reproducible. The added archetype analysis further shows that the context signatures support an interpretable scene taxonomy rather than only frame-level verification. This provides a solid basis for future retrieval, cross-dataset archetype transfer, or predictive use of cut-in scenarios.

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Appendix A

Appendix: Terminology and highD field mapping

To maintain the focus of the main chapters on concepts rather than code-level details, we describe lanes and neighbour relations in descriptive terms such as *same-lane leader* or *right-lane follower*. Table A.1 lists the corresponding column names in the released highD dataset for readers who want to reproduce the evaluation directly from the raw files.

Table A.1: Mapping between thesis terminology and highD columns (tracks file).

Concept in this thesis	highD column	Meaning
Lane index (reference labels)	`laneId`	Discrete lane identifier per frame.
Same-lane leader (ahead)	`precedingId`	Track id of closest vehicle ahead in the same lane (0 if none).
Same-lane follower (behind)	`followingId`	Track id of closest vehicle behind in the same lane (0 if none).
Left-lane leader	`leftPrecedingId`	Closest vehicle ahead in the left adjacent lane (0 if none).
Left-lane alongside	`leftAlongsideId`	Vehicle alongside in the left adjacent lane (0 if none).
Left-lane follower	`leftFollowingId`	Closest vehicle behind in the left adjacent lane (0 if none).
Right-lane leader	`rightPrecedingId`	Closest vehicle ahead in the right adjacent lane (0 if none).
Right-lane alongside	`rightAlongsideId`	Vehicle alongside in the right adjacent lane (0 if none).
Right-lane follower	`rightFollowingId`	Closest vehicle behind in the right adjacent lane (0 if none).

HighD distributes each recording as three CSV files:

- `XX_tracks.csv` (per-frame trajectories),

- `XX_tracksMeta.csv` (per-track metadata),
- `XX_recordingMeta.csv` (recording metadata, including lane markings).

Appendix B

Appendix: Plain-language algorithms

B.1 Lane change detection (stable blocks)

1. For each vehicle, read its lane id over time (reference) or inferred lane index (XY-lane).

2. When the lane id changes from L_1 to L_2, check:

   - the vehicle stayed in L_1 for at least N_{before} frames before the change,
   - the vehicle stayed in L_2 for at least N_{after} frames after the change.

3. If both checks pass, emit a lane change event from L_1 to L_2.

B.2 Cut-in detection (follower identification)

1. Start from a lane change event where the cutter enters a target lane.

2. Move forward frame by frame within a configured search window.

3. In each frame, look for any vehicle in the target lane whose same-lane predecessor equals the cutter id.

4. When this first happens, that vehicle is the follower and the frame is t_0.

5. Require that the follower relation holds for a minimum number of frames to reduce noise.

B.3 DHW, THW, and TTC for the interacting pair

- DHW is computed as the longitudinal gap between the cutter's rear and the follower's front.
- THW is DHW divided by follower speed (with safe handling of near-zero speeds).
- TTC is DHW divided by closing speed when the follower is gaining on the cutter; otherwise TTC is treated as infinite.

B.4 Reference 3×3 occupancy grid from highD neighbour ids

For a given ego vehicle and frame, highD provides surrounding-vehicle ids for left/right/preceding/following/alongside positions. We map these to the 3×3 grid by setting a cell to 1 if the corresponding neighbour id is non-zero; otherwise 0. The same grid can be produced by geometry-based reconstruction, enabling code-level verification via SFC signature matching.

Appendix C

Appendix: Supplementary occupancy maps

Figure 7.1 in Chapter 7 presents the decision-stage occupancy maps in the main text because it provides a representative pre-onset view of the surrounding traffic configuration. For completeness, Figure C.1 reports additional absolute occupancy maps for the intention and execution stages.

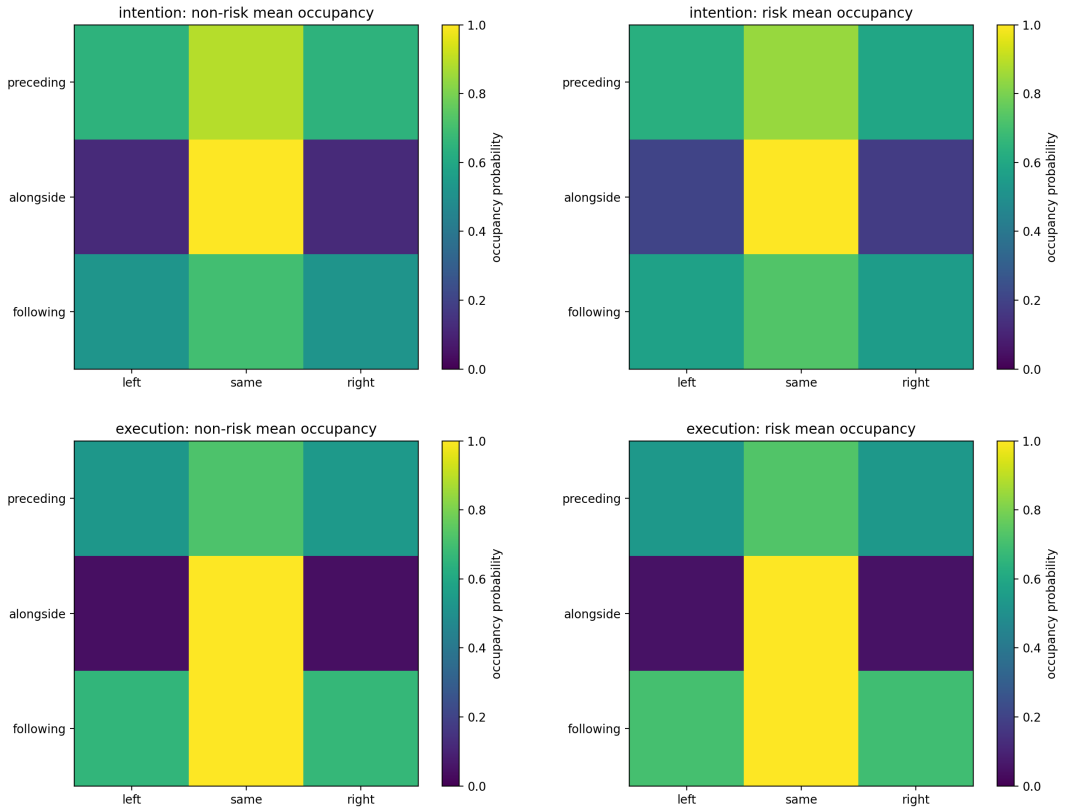

Figure C.1: Supplementary absolute occupancy maps for the intention and execution stages (non-risk vs risky). These plots complement the decision-stage example in Figure 7.1.